

Q5: Student Biological & Statistical Study Guide

A comprehensive, student-focused reference explaining the **biology**, **statistics**, **machine learning**, and **dataset logic** behind Question 5.

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1. The Clinical Problem & Project Goal

In cutaneous melanoma, patients diagnosed with advanced (Stage III/IV) disease have three main treatment avenues:

- Arm A — Immunotherapy:** Immune Checkpoint Inhibitors (ICI) such as Anti-PD-1 (*Nivolumab*, *Pembrolizumab*) or Anti-CTLA-4 (*Ipilimumab*). These drugs release the molecular "brakes" on cytotoxic CD8+ T cells so they can destroy tumour cells.
- Arm B — Targeted Therapy:** Small-molecule kinase inhibitors (BRAF inhibitor *Dabrafenib* + MEK inhibitor *Trametinib*). These selectively block hyperactive MAPK pathway signaling in tumours harboring a **BRAF V600E/K mutation**.
- Arm C — Chemotherapy & Combination Therapy:** Cytotoxic chemotherapy (*Dacarbazine*) or combination regimens (e.g., M2-macrophage depletion / epigenetic priming combined with anti-PD-1).

The Challenge

If immunotherapy is given indiscriminately to all patients, only ~35–40% respond. For non-responders, time is wasted while the tumour progresses, and patients experience severe immune-related adverse events.

The Q5 Objective

Question 5 builds a **biomarker-guided decision support system**. By inspecting a patient's baseline molecular profile (gene expression signatures, M1/M2 ratio, driver mutations like *BRAF*, *NRAS*, *NF1*, and mutational burden), the pipeline determines whether the patient will respond to Immunotherapy (Arm A), should switch to Targeted Therapy (Arm B), or requires Combination/Chemotherapy (Arm C).

2. Biological Concepts Explained

Immune Checkpoints & T-Cell Infiltration

- CD8+ Cytotoxic T Cells:** The primary immune effector cells capable of killing cancer cells.
- PD-1 (*PDCD1*) & PD-L1 (*CD274*):** PD-1 is a receptor expressed on T cells. Tumours express PD-L1 (*CD274*), which binds to PD-1 (*PDCD1*) and sends an inhibitory "don't kill me" signal to the T cell. Anti-PD-1 antibodies block this binding, restoring T-cell cytotoxicity.
- Tumour Inflammation Signature (TIS):** An 18-gene expression signature measuring pre-existing adaptive immune suppression and active IFN- γ signaling in the tumour microenvironment.
- Cytolytic Index (CYT):** Defined as the geometric mean of Perforin 1 (*PRF1*) and Granzyme A (*GZMA*) transcript levels, quantifying active cytotoxic T-cell and NK-cell killing.

Macrophage Polarisation (M1 vs. M2) & Macrophage STV

Tumour-Associated Macrophages (TAMs) exist along a functional spectrum:

- M1 Macrophages (Pro-inflammatory / Antitumour):** Produce pro-inflammatory cytokines (*TNF*, *IL12B*, *CXCL10*) and assist T cells in tumour destruction.
- M2 Macrophages (Anti-inflammatory / Pro-tumour):** Produce immunosuppressive factors (*IL10*, *TGF β 1*, *ARG1*), stimulate angiogenesis, and recruit regulatory T cells (Tregs), blocking anti-PD-1 efficacy.

The Macrophage STV Matrix (*m1_m2_stv.csv*)

The project utilizes a linear **Signature Transcript Vector (STV)** containing 14,837 gene weights (W_g). For patient i with $\log_2$ -transformed expression $E_{i,g}$:

$$\text{STV Score}_i = \sum_{g=1}^{14837} W_g \cdot E_{i,g}$$

- Positive weights correspond to M1-associated features; negative weights correspond to M2-associated features.
- The normalized score yields the **M1/M2 Ratio** ($M1/[M1 + M2]$). High ratios reflect an inflamed, M1-dominant microenvironment; low ratios indicate M2-mediated immune exclusion.

Mechanisms of Immunotherapy Resistance

Non-response to checkpoint blockade typically stems from four biological barriers:

1. **Antigen Presentation Defect:** Loss-of-function mutations or silencing in β_2 -microglobulin (B2M) or transporter genes (TAP1 , TAP2 , HLA-A , HLA-B , HLA-C) prevent HLA class I molecules from displaying tumour neoantigens on the cell surface.
2. **IFN- γ Pathway Resistance:** Loss of JAK1 , JAK2 , STAT1 , or STAT3 prevents tumour cells from responding to interferon-gamma, making them insensitive to T-cell killing.
3. **Immunosuppressive Stroma:** High expression of IDO1 , TGF β 1 , or ARG1 creates a metabolic barrier that depletes essential amino acids (tryptophan, arginine) needed for T-cell survival.
4. **Copy Number Alteration (CNA) Burden & Proxy Strategy:** High chromosomal instability / aneuploidy physically disrupts T-cell infiltration.

Note

Methodological Choice: iAtlas Harmonisation vs. CNA Availability: Unharmonised cBioPortal downloads for trial cohorts (Liu 2019, Hugo 2016, Riaz 2017) do not contain GISTIC CNA files. Attempting to use unharmonised RNA-seq to gain CNA for TCGA alone would introduce severe cross-cohort batch artifacts. By using iAtlas harmonised expression data, cross-cohort batch effects are eliminated. Chromosomal instability is represented via WES-derived Tumour Mutational Burden (TMB), while gene loss (PTEN , CDKN2A) is captured via combined mutation status and transcript abundance.

3. Statistical & Machine Learning Concepts Explained

Univariate Association Testing & Youden Cutoffs

To evaluate individual biomarkers (X) against binary response ($Y \in \{0, 1\}$):

- **Mann-Whitney U Test (Continuous Features):** A non-parametric test assessing whether the distribution of signature scores (e.g. TIS, CYT) differs significantly between Responders and Non-Responders without assuming normal distributions.
- **Cohen's d (Effect Size):** Quantifies the standardized mean difference between groups:

$$d = \frac{\bar{X}_1 - \bar{X}_0}{s_{\text{pooled}}}$$

Values of $d > 0.8$ indicate large biological effect sizes.

- **Fisher's Exact Test (Categorical Features):** Used for contingency tables (e.g., BRAF mutation status vs. Response) to calculate exact p -values, Odds Ratios (OR), and 95% confidence intervals.
- **Youden's J Statistic (Optimal Threshold):** Identifies the numeric cutoff c^* on a continuous feature that maximizes diagnostic accuracy:

$$J(c) = \text{Sensitivity}(c) + \text{Specificity}(c) - 1$$

The cutoff c^* corresponding to $\max J(c)$ is used to discretize continuous signatures into clinical high/low categories.

Logistic Regression Interaction Terms

To test whether two features act synergistically:

$$\text{logit}(P(Y = 1)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \cdot X_2)$$

A statistically significant interaction coefficient ($\beta_3 \neq 0, p < 0.05$) indicates that the predictive effect of feature X_1 (e.g., TIS score) depends on the status of feature X_2 (e.g., BRAF mutation status).

Unsupervised Clustering

Feature Normalization (StandardScaler)

Before clustering, continuous features are standardized to zero mean and unit variance ($z = \frac{x - \mu}{\sigma}$). This prevents features with large raw scales (e.g. TMB) from dominating distance calculations over bounded signature scores.

K-Means vs. Hierarchical Agglomerative (Ward) Clustering

- **K-Means:** Iteratively assigns patients to K clusters by minimizing total within-cluster variance (inertia).
- **Agglomerative (Ward Linkage):** A bottom-up hierarchical approach that merges pairs of clusters to minimize the increase in total within-cluster variance at each step.

Cluster Validation Metrics

- **Silhouette Coefficient (s):** For patient i , measures how similar i is to its own cluster (a_i) compared to the nearest neighboring cluster (b_i):

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad s_i \in [-1, 1]$$

- Mean silhouette score > 0.5 indicates strong cluster structure.
- **GAP Statistic:** Compares the total within-cluster variation W_k against expected null values W_k^* generated by Monte Carlo sampling from a uniform distribution. The optimal K maximizes $\text{Gap}(K) = E^*[\log(W_k)] - \log(W_k)$.
 - **UMAP (Uniform Manifold Approximation and Projection):** A non-linear dimensionality reduction technique that projects high-dimensional signature space down to 2D for visual inspection while preserving local and global data topology.

Decision Curve Analysis (DCA), NNT & PPV

Evaluating a model using AUC-ROC alone is insufficient for clinical adoption because AUC does not account for the real-world clinical costs of false positives vs. false negatives.

Net Benefit Formula

For a decision threshold probability $p_t \in [0, 1]$ (the probability threshold at which a doctor recommends treatment):

$$\text{Net Benefit} = \frac{\text{True Positives}}{N} - \left(\frac{\text{False Positives}}{N} \right) \times \left(\frac{p_t}{1 - p_t} \right)$$

- **Interpretation:** If a model's DCA curve lies above both the "Treat All" and "Treat None" strategies across relevant thresholds ($p_t = 0.2 - 0.6$), using the model to guide treatment decisions adds net clinical benefit.

Clinical Utility Metrics

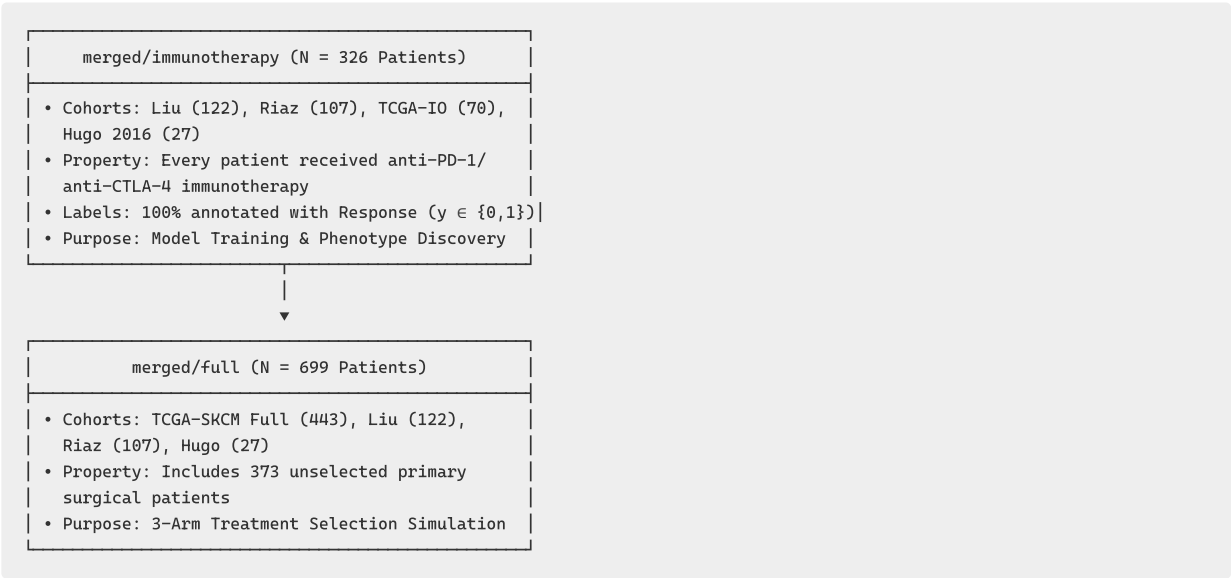
- **Number Needed to Treat (NNT):** The number of patients who must be treated according to the model to achieve one additional positive response:

$$\text{NNT} = \frac{1}{\text{Absolute Risk Reduction}}$$

- **Positive Predictive Value (PPV):** The probability that a patient predicted as "Responder" actually achieves clinical response (CR/PR).

4. Dataset Strategy: merged/immunotherapy vs. merged/full

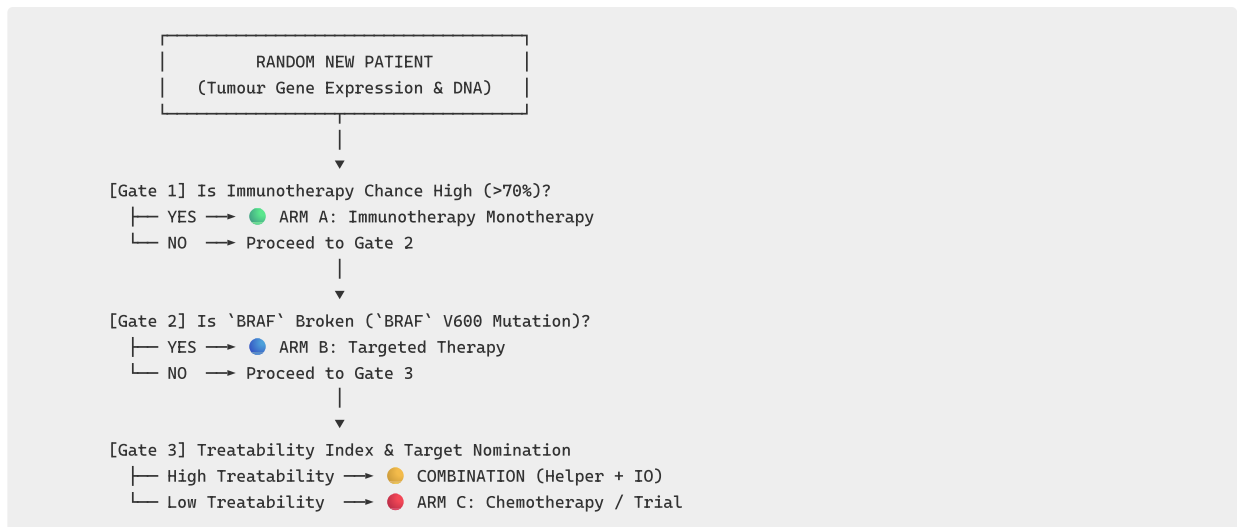
Understanding why we use two merged datasets resolves common confusion:



1. **Why Train on merged/immunotherapy (N = 326):**
To fit machine learning classifiers ($\hat{y} = f(X)$) or identify response-associated phenotypes, we must have ground-truth response labels (y). merged/immunotherapy is the only dataset where 100% of patients have verified immunotherapy outcomes.
2. **Why Deploy on merged/full (N = 699):**
When a random new patient arrives at the clinic, we do not know if they will respond to immunotherapy. merged/full represents the general, unselected melanoma population where Q5's 3-arm decision tree assigns patients to Immunotherapy, Targeted Therapy, or Chemotherapy.

5. The 3-Arm Decision Tree Architecture

The master decision engine routes patients through three sequential evaluation gates:



Summary of Script Mapping

- [scripts/01_load_and_prepare.py](#): Loads data, extracts signatures, applies Macrophage STV, computes deconvolution.
- [scripts/02_feature_analysis.py](#): Runs MW-U, Fisher's, Youden cutoffs, and interaction tests.
- [scripts/03_cluster_patients.py](#): Performs K-Means/Ward scaling, silhouette/GAP evaluation, and UMAP projection.
- [scripts/04_phenotype_characterisation.py](#): Annotates phenotypes (*Immune Hot*, *Cold*, *M2-High*, *Mutant-Driven*), KM survival, and Q3 ODE trajectories.
- [scripts/05_subgroup_models.py](#): Trains subgroup-specific models using LOCO cross-validation.
- [scripts/06_clinical_utility.py](#): Computes DCA net benefit curves, NNT, and PPV.
- [scripts/07_treatability_scoring.py](#): Scores treatability index, maps Q4 DepMap/LINCS drug targets, and constructs the 3-arm decision tree.